CAMERA MODULE TECHNICAL DATASHEET

Model: [TS2M3V] **Revision:** [V1.0]

1. Product Overview

PRODUCT IMAGE	
	Model: **TS2M3V** Image Sensor: **OV2735** Resolution: **2MP** Interface: **USB 2.0** Frame Rate: **1080P @ 30fps** Module Size: **80 × 16 mm**

Product Description
The TS2M3V is a 2MP USB 2.0 camera module designed for integration into electronic imaging devices and embedded vision systems. It supports multiple output resolutions, including 640 × 480, 800 × 600, 1280 × 720, and 1920 × 1080, with YUV and MJPG video formats. The module features a 76° field of view and USB bus power for system integration.

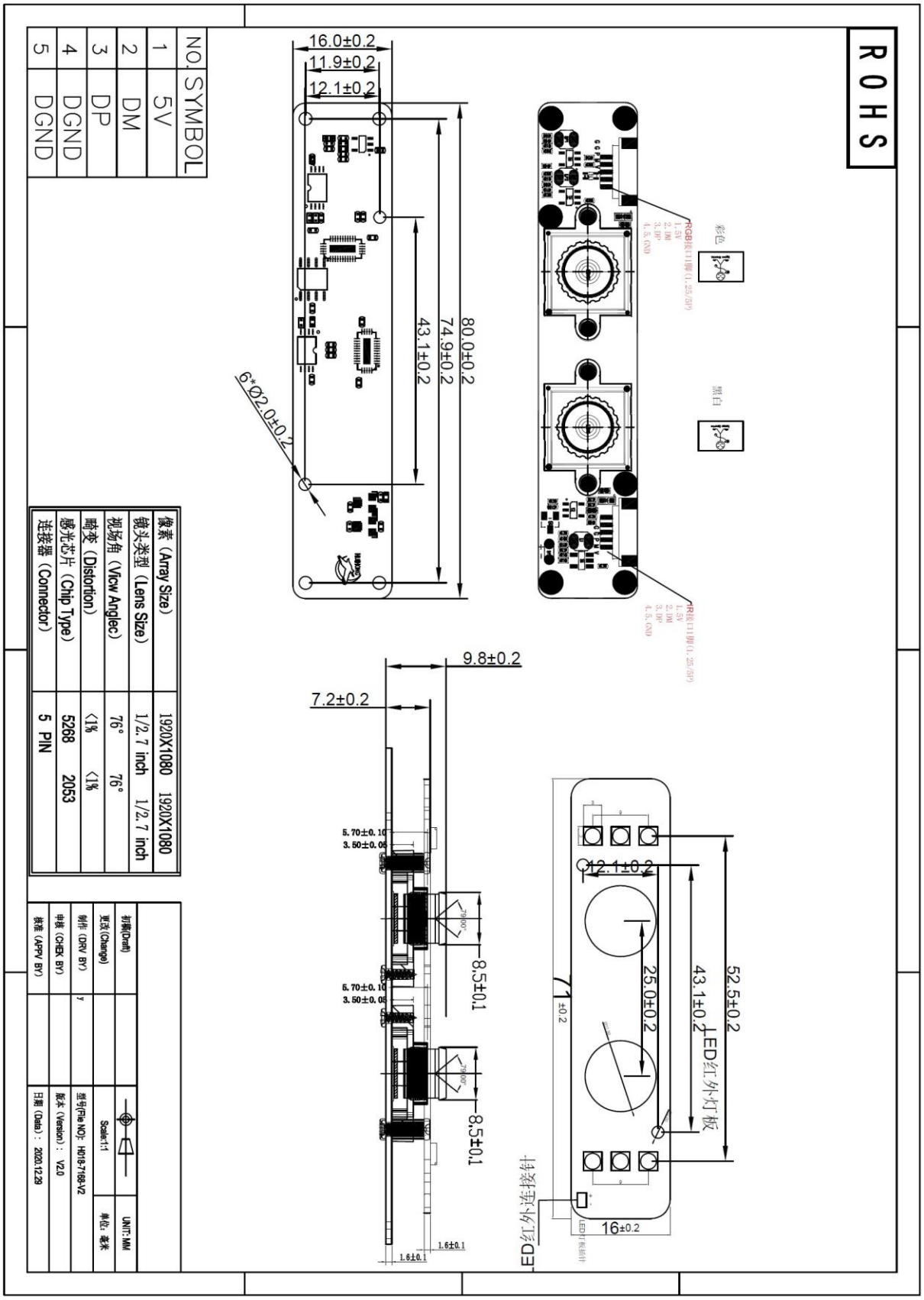
2. Key Features

- **2MP Imaging:** Supports 1920 × 1080 image output.
- **Multiple Resolutions:** Supports 640 × 480, 800 × 600, 1280 × 720, and 1920 × 1080.
- **USB 2.0 Interface:** USB bus-powered design.
- **Compact Design:** Module dimensions of 80 × 16 mm.
- **Optical Specifications:** 76° field of view and F/2.0 lens aperture.
- **Video Formats:** Supports YUV and MJPG compression formats.

3. Technical Specifications

Item	Specification
Model	TS2M3V
Image Sensor	OV2735
Sensor Type	1/2.7-inch
Resolution	1920 × 1080 (2MP)
Pixel Size	3.0 × 3.0 μm
Video Output	YUV / MJPG
Maximum Frame Rate	30Fps
Lens	4.35 mm, F/2.0
Field of View	76°
Focus	FF, AF
WDR / HDR	Dynamic range: 85 dB
Interface	USB 2.0
Operating Voltage	USB bus power; voltage not specified
Operating Current	400 mA
Operating Temperature	-20°C to +60°C
Stable Image Temperature	0°C to +50°C
Module Dimensions	80 × 16 mm (L × W); height not specified
OS Compatibility	Windows XP, Windows Vista, Windows 7

4. Mechanical Dimensions



Unit: mm. Dimensions are for reference. Please refer to the latest approved mechanical drawing for production.

Module Dimensions: 30 × 58 × 16 mm

5. Interface & Pin Definition

Pin	Name	Type	Function Description
1	VDD	Power	USB power supply +5V
2	DM	Data Pin	USB data transmission (D−)
3	DP	Data Pin	USB data transmission (D+)
4	GND	Ground	System ground
5	GND	Ground	System ground

6. Notes

- Specifications are subject to change without prior notice.
- Actual imaging performance may vary depending on lighting conditions and host platform.
- Operating temperature and stable-image temperature are specified separately.
- The module uses USB bus power and has a specified operating current of 400 mA.
- Handle the module using appropriate electrostatic discharge (ESD) precautions.
- Avoid bending the PCB or FPC during storage, transportation, assembly, disassembly, and testing.
- Do not connect or disconnect the module while powered on.
- Ensure that the cable is connected in the correct orientation and pin alignment.
- Please contact TUXIVISION for the latest specification and mechanical drawing.